

From Zurich to Valais: Phonetic Variation and its Impact on Forensic Automatic Speaker Recognition

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Abstract

Within the German-speaking world, Switzerland stands out for its sustained and socially prestigious use of dialects whose persistence, along with their considerable variation in phonetic, phonological, and lexical features, poses both opportunities and challenges for phonetic research and speech-processing systems. This study examines how dialectal variation within Swiss German affects the performance of automatic speaker recognition (ASR) systems. Although automatic speaker recognition systems are often considered language-independent, practical evaluations involving less widely spoken languages, such as Swiss German, or distinct dialect groups remain limited. In particular, the impact of dialect mismatches between reference datasets and test samples during likelihood estimation is not yet well understood, leaving many open questions. Understanding these effects is essential, particularly in applied contexts such as forensic phonetic casework. This practical relevance provided the motivation for the present study. Developed in collaboration with the Forensic Institute Zurich (FOR), this study aims to provide insight into reference data selection in contexts of low resource varieties such as Swiss German. Using speech data from the SDATS corpus (Swiss German Dialects Across Time and Space), ASR performance across four dialect regions (Zurich, Bern, Valais, Graubünden), based on recordings from 102 male speakers, was analysed. The dataset includes samples that vary in speaking style, degree of dialectality, and recording quality, closely resembling realistic forensic conditions. Preliminary results suggest good overall system performance, but subtle differences depending on the inclusion of standard Swiss German speech in the comparisons. Findings align with previous work and highlight the practical value of insights into phonetic variation when working with ASR systems.

Keywords: dialectal variation, Swiss German, phonetic diversity, ASR, reference data

1. Introduction and background

Language use in Switzerland represents a notable exception within the German-speaking world. While the standard variety is primarily reserved for formal and written contexts, dialects constitute the unmarked, default mode of communication in everyday life. They are highly prestigious, central to identity construction, and used with a matter-of-factness rarely observed in other German-speaking countries (Herrgen 2019: 246). This special form of diglossia results in few fluent transitions between dialect and standard varieties (Ammon et al. 2018: LXV), although exceptions in certain sociolinguistic contexts (Ruch et al. 2023) and due to language change (Leemann et al. 2024; Graf 2015) do exist. Unlike in Germany, there is no comparable shift toward regionally coloured standard varieties, and the Swiss dialect landscape remains relatively robust (Ruch et al. 2023).

Ultimately, this results in a high degree of phonetic diversity within Swiss German. Swiss German dialects can be grouped into three major Alemannic varietal groups, yet significant variation also exists along west–east and north–south axes. These varieties differ along several

phonetic dimensions, e.g. in vowel quality and quantity, consonant realizations, intonation and speech rate (Leemann et al. 2024; Ruch et al. 2023).

This diversity might pose a challenge for automatic speaker recognition (ASR) systems. In order to effectively apply ASR in forensic phonetic casework in German-speaking Switzerland—or in any low-resource dialect region—the interchangeability of reference corpora becomes a crucial issue. It must be assessed whether a general corpus of Swiss Standard German provides sufficient reference material, or whether dialect-matched corpora yield improved performance.

2. Methodology

This study examines the extent to which regional phonetic variation influences the performance of ASR systems. Focusing on four Swiss German dialect regions, it evaluates the effects of dialectal differences (dialect vs. standard), speaking style (read vs. spontaneous), and recording conditions (different types of background noise) that resemble forensic scenarios.

The speaker data stem from the SDATS corpus (Leemann et al., 2020) and include recordings from 102 male speakers aged 17 to 43, distributed across the four dialect regions (Zurich: $n = 24$, Bern: $n = 32$, Valais: $n = 22$, Graubünden: $n = 24$). While we refer to four regional groups, it should be noted that these categories are a simplification: dialect boundaries are dynamic and shaped by factors such as age, locality, education, and sociolinguistic context (Leemann et al., 2024; Steiner et al., 2023). For each speaker, three recordings were selected, representing distinct speech modalities: read speech in Swiss Standard German (~25 sec.), read speech in dialect (~2 min.) and spontaneous speech (~1 min.). Due to COVID-19 constraints, SDATS data were collected remotely via a supervised smartphone app, leading to variable audio quality resembling forensic conditions. To account for these factors, a noise classification scheme based on Jessen (2012) was developed and applied during annotation. The categorized noise types and their distribution will be presented at the conference as part of a broader diagnostic assessment of potential confounding variables.

Testing was conducted using the VOCALISE 2021, which is a state-of-the-art software program for performing speaker comparisons using Deep Neural Network (DNN) based x-vectors. Performance evaluation was done with the complementary software Bio-Metrics 2024. The values reported correspond to standard performance metrics commonly used in ASR evaluation, namely equal error rate (EER, the value when false rejection and false acceptance rate are identical) and the log-likelihood-ratio cost (Cllr, which reflects the accuracy of an ASR recognition system). Two testing approaches were used. First, the performance of the automatic speaker recognition system Vocalise was evaluated quantitatively using EER and Cllr metrics. Second, the impact of different reference databases on the resulting likelihood ratios (LRs, not log-transformed) was examined qualitatively.

3. Preliminary results

Initial results after cross-validation indicate strong baseline ASR performance, with EERs between 0.4% and 1.3% (convex hull) and Cllr values from 0.02 to 0.09. Slightly lower values (indicating even better performance) were observed when Swiss Standard language samples were excluded from the reference data, possibly reflecting effects of the diglossia in German speaking Switzerland—though this requires further investigation. Initial 1-vs.-1 comparisons using varying reference sets also produced very high likelihood ratios. Taken together, these preliminary findings suggest that the ASR system performs well on dialectal speech data, in line with previous findings (Watt et al. 2020). The limited number of speakers meant that calibration aspects could only be addressed to a restricted extent. Nevertheless, observations of outlier speakers suggest that calibration should be a key consideration in future studies. Further analyses on how dialectal variation affects ASR outcomes across different dialect groups and

speech modalities will be presented, focusing on additional influencing factors such as background noise and speaking style.

4. References

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